

Ara Yeroyan

SENIOR AI ENGINEER · LLMS, GENAI, VOICE AI · AGENTIC AI & RAG

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Senior AI Engineer with **6+ years** building end-to-end ML and GenAI across **voice, retrieval, and applied NLP**, including published work with the **NVIDIA NeMo team**. I design, evaluate, and deploy systems that must hold up in production: LLM-powered products, voice agents, RAG pipelines, multi-agent orchestration, and the evaluation/observability that proves they work. Peer-reviewed at SIGIR and INTERSPEECH; Google Summer of Code mentor.

PROFESSIONAL EXPERIENCE

ServiceTitan Sep 2024 - Present
Senior AI Engineer, Research · Yerevan

- LLM evaluation & monitoring pipelines for voice agents, quality, safety, drift, regressions.
- Optimized LLM inference for a **3.2× performance improvement**; built LLM-based reporting over high-volume operational data.
- Defined and operationalized customer-facing KPIs tied to revenue.

Akryl Digital (Client: GIZ) Aug 2025 - Apr 2026
ML Architect · Consultant · Hamburg (remote)

- Production multi-agent chatbot systems for GIZ, RAG pipelines at **sub-2-second latency**; primary technical liaison.
- Built visual RAG transforming PDFs into high-resolution images with multi-vector retrieval and token-level filtering.

Plat.ai Sep 2022 - Mar 2025
Senior ML Engineer · Consultant · Yerevan

- Built Plat.ai's AutoML and AdvancedML products with built-in explainability and interpretability.
- Domain-specific features for financial ML; **+17% F1** for banking and insurance clients.

yess.ai (YC-backed) Jan - Jun 2024
Senior NLP Engineer · Palo Alto (remote)

- Real-time voice assistant handling live calls with external tools for appointment booking/cancellation.
- Offline sales-assistant workflows using tree-of-thought and self-reflection agent patterns.

EPAM Systems Jun - Sep 2024
Senior ML Engineer (enterprise retail/fashion) · Yerevan

- Real-time multimodal recommendation MVP ("Fashion Assistant") for a major enterprise client.

Podcastle Jul 2021 - Aug 2022
Machine Learning Specialist · Yerevan

- Audio synthesis & enhancement, voice cloning, DSP; led data-collection and training/serving pipelines.

Earlier 2020 - 2024
Lecturer · Armenian Code Academy · ML at Funday, FAST Foundation / CYNORA, Hero House

PUBLICATIONS

ORCID 0009-0009-3605-9702 · Google Scholar d8If8-gAAAAJ

SIGIR 2026 · ACCEPTED

Visual RAG Toolkit: Scaling Multi-Vector Visual Retrieval with Training-Free Pooling and Multi-Stage Search
Yeroyan, A. · doi.org/10.1145/3805712.3808383 · arXiv:2602.12510

INTER_SPEECH 2025 · WITH NVIDIA NEMO
From Scarcity to Sufficiency: A Speech-Recognition Pipeline for Zero-Resource Languages
Karpov, N., Kostandian, S., ... Yeroyan, A., et al.

PREPRINT · ARXIV, JAN 2025
Methods to Increase the Amount of Data for Speech Recognition for Low-Resource Languages
Ayrapetyan, A., Kostandian, S., Yeroyan, A., et al. · arXiv:2501.14788

PREPRINT · ARXIV, JUN 2024
Enabling ASR for Low-Resource Languages: A Comprehensive Dataset Creation Approach
Yeroyan, A., Karpov, N., et al. · arXiv:2406.01446

EDUCATION

MSBA, UCLA (Anderson) Class of 2027
STEM-designated · from Sept 2026

BSc Data Science, American University of Armenia 2020 - '24
GPA 3.85 / 4.00 · top 1% of cohort

HONORS & AWARDS

Presidential Award, Best Bachelor Student in the field of IT in Armenia 2024
Republic of Armenia & Synopsys · head-of-state recognition

Silicon Valley Exchange Track 2023
Palo Alto / Sunnyvale, California

Google Summer of Code, Mentor (DBpedia) 2026
GMAT 720

KEY EXPERTISE

LLMs · GenAI · Agentic AI · RAG & Retrieval · Voice AI & Applied NLP · LLM Eval & Observability · Multi-agent orchestration · MLOps · Low-resource speech / ASR · Python

LANGUAGES

Armenian, native · English, C1 · Russian, C1